

# Kai Barker

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## EDUCATION

### University of California, Santa Barbara

Bachelor of Science (B.S.), Statistics and Data Science

September 2022 - June 2026

Santa Barbara, CA

Major GPA: 3.90/4.0

Activities: Consult Your Community, Statistics & Data Science Club, Data Science Collaborative, Ski & Snow Club

Coursework: Linear Algebra, Optimization, Machine Learning, Design of Experiments, Time Series, Adv. Statistical Models, Big Data

## SKILLS & CERTIFICATES

Languages & Tools: SQL, Python (pandas, NumPy), R (tidyverse, dplyr), Microsoft Excel, PostgreSQL, Git, LaTeX, API Integration, HTML/CSS

Analytics & Visualization: Tableau, Plotly, Seaborn, Matplotlib, ggplot2, Hypothesis Testing, PowerPoint, Interactive Dashboards

Machine Learning & Statistics: Scikit-Learn, XGBoost, PCA, Feature Selection (LASSO, Ridge), Bootstrap/Permutation Testing

Big Data & Engineering: Apache Spark, PySpark, GCP (BigQuery, Dataproc, GCS), ETL Pipelines, Distributed SQL, Parquet

Certificates: HackerRank SQL (2026), Anthropic AI Fluency Framework & Foundations (2026), JP Morgan Chase SWE (2025)

## EXPERIENCE

### Data and Systems Consultant

June 2026 - Present

Santa Barbara Wildlife Care Network

Santa Barbara, CA

- Identified 16-20 hours of weekly coordination overhead and \$1,000+/year in software costs through volunteer experience and stakeholder interviews; engaged as outside consultant to design and build a replacement system for 250+ volunteers.
- Built a PostgreSQL backend and React frontend with automated check-in, shift management, and onboarding workflows, projected to reduce manual coordination overhead to near zero and cut software costs to under \$14/month.

### Engagement Manager and Business Analyst

October 2025 - June 2026

Consult Your Community

Santa Barbara, CA

- Led a 16-week EdTech market-entry engagement as Engagement Manager, directing a 7-person team across 3 workstreams toward the client's first paid B2B pilot.
- Modeled \$26M in recoverable funding across 115 CA community colleges by building an Excel financial model and Tableau dashboards, communicating findings directly to non-technical clients to drive recommendations.
- Translated data findings, including that 70% of corporate clients had lapsed within a year, into actionable deliverables: a CRM system, automated email reactivation campaigns, and a buyer-ready SOP for acquisition preparation.
- Identified \$40,000 in recoverable annual revenue by extracting and merging transactional data from Shopify, Square, and client records, then segmenting corporate accounts by order value, activity, and lifetime value.

### Data Science Researcher

September 2025 - June 2026

Department of Statistics and Applied Probability | Advised by Department Chair

Santa Barbara, CA

- Investigated whether congressional stock trades carry predictive signal, building an end-to-end Python pipeline to clean and reconcile 137,000+ disclosure records into 15,000+ purchase events and measure CAPM-adjusted abnormal returns against the S&P 500.
- Identified a statistically significant signal in a subset of trades, top-performing members and coordinated purchases generated abnormal 30-day returns of +1.4% while unfiltered trades showed none, distinguishing a genuine signal from a tradable strategy.
- Authored a research paper and presented findings to the department chair, applying Random Matrix Theory covariance denoising and convex portfolio optimization, and documenting results with full visualizations and a limitations analysis.

### Research Assistant

September 2024 - July 2026

Foundations of Financial Technology Lab

Santa Barbara, CA

- Quantified extreme wealth concentration in Ethereum (Gini coefficient of 0.96) by building a pipeline to process 20M+ blocks and 100M+ wallets across a 5-year period, showing the top 1% of accounts control 95%+ of Ether supply.
- Improved accuracy of distribution metrics by filtering 65,000+ institutional and artificial accounts to isolate individual holdings.
- Credited for analysis contributions to a published blockchain inequality paper alongside researchers from Yale, Cornell, and UCSB.

## PROJECTS

### Gene Therapy Off-Target Effect Analysis

May 2026

- Engineered an end-to-end ETL pipeline on GCP using PySpark, Dataproc, and BigQuery SQL to process a 78,893-gene expression matrix and 705,000+ disease association records across 22 candidate gene therapy treatments for Angelman syndrome research.
- Determined the single safest of 22 candidate treatments by building a distributed validation framework that ran 1,800+ model fits in parallel and flagged statistically significant off-target effects ( $p < 0.001$ ).

### NeurIPS Ariel Data Challenge

July 2025

- Designed a signal processing pipeline to calibrate 200+ GB of raw telescope data across 14,000 files and 1,100+ exoplanets, correcting detector noise and instrument distortion to recover usable atmospheric signal for the European Space Agency's Ariel mission.
- Constructed a weighted ensemble of Gaussian Process Regression, XGBoost, and Ridge models to predict atmospheric composition across 283 wavelength channels, tuning uncertainty estimates via grid search to optimize Gaussian Log Likelihood.